SOIL EROSION AND CONSERVATION · NOTE ST02

Hydraulics of Conservation Structures — Froude Number, Critical Depth, Hydraulic Jump and Stilling Basins

Course topics: SWCE-503 T6, SWCE-503 T8

Froude number

$$F_n = V/\sqrt{(gL)}\ \ L = \text{hydraulic depth}\ D = a/w\ (a = \text{flow area}, w = \text{surface width}; D = \text{depth for a rectangle})$$

F_n is the dimensionless ratio of inertial to gravity forces. $F_n = 1$ ($V = \sqrt{(gD)}$) is **critical flow**.

Flow	Condition	Character
Subcritical (tranquil, streaming)	$F_n < 1$, $V < \sqrt{(gD)}$	Gravity dominant, low velocity
Supercritical (rapid, shooting, torrential)	$F_n > 1$, $V > \sqrt{(gD)}$	Inertia dominant, high velocity

Specific energy and critical depth

$$\text{Specific energy: } H_e = y + V^2/2g$$
$$\text{With } q = a{\cdot}V \text{ and } a = b{\cdot}y \text{ (rectangle): } H_e = y + q^2/(2a^2g) = y + q^2/(2b^2y^2g)$$
$$\text{Critical depth: } d_c = \sqrt[3]{(q^2/(b^2g))}\ \text{At } y = d_c\text{: } H_e = (3/2){\cdot}y$$

- **Specific energy** is the energy per unit weight measured from the channel bottom — depth plus velocity head.
- **Critical depth** is the depth at which discharge is maximum for a given total head (depth + velocity head), or equivalently where total energy is minimum. **Critical flow** occurs where the slope just maintains critical depth.

Hydraulic jump

- A sudden, turbulent rise from below critical depth to above it — the flow changes from supercritical to subcritical, kinetic energy turns into potential energy, and turbulence dissipates energy. The water surface rises abruptly, velocity falls and depth increases from d_1 to d_2 ($d_1 < d_2$). A **local phenomenon** is a rapid change of flow state over a short distance — a hydraulic jump or a hydraulic drop.
- **Uses:** dissipate the energy of water falling over dams, weirs and structures to prevent downstream scour; recover head or raise the downstream level for irrigation canals and distribution; add weight of water to cut uplift under masonry structures.
- **Height of jump** = $d_2 - d_1$. Jump formation depends on the kinetic-flow factor $V^2/(gd)$, the critical depth and the Froude number $V/\sqrt{(gd)}$. At $F_n = 1$ the flow is at critical depth and no jump forms.

$$\text{Sequent depth ratio: } d_2/d_1 = ½{\cdot}(\sqrt{(1 + 8F_n{}^2)} - 1)$$

Jump type	F_n	Character
Undular	1 to 1.7	Undulating surface
Weak	1.7 to 2.5	Small surface rollers; smooth downstream; low energy loss
Oscillating	2.5 to 4.5	Oscillating jet moving bottom to surface without periodicity; waves travel beyond the basin
Steady	4.5 to 9	Stable, well balanced; turbulence confined; fairly smooth downstream
Strong	> 9	High turbulence, growing rollers, rough downstream surface

Illustration (not in the notes)
$F_n = 4$, $d_1 = 0.2$ m → $d_2/d_1 = ½(\sqrt{(1 + 128)} - 1) = ½(11.36 - 1) = 5.18$ → $d_2 \approx 1.04$ m (a steady jump)

Stilling basins

- A **hydraulic-jump basin (dissipator)** is a paved stilling basin below a spillway, chute or drop — at spillway feet, canal structures, culvert outlets and transitions — where high-velocity supercritical flow forms an efficient jump, reducing velocity and energy so the channel does not erode.
- Appurtenances add roughness, dissipating more energy and shortening the basin. For straight-inlet drops, energy is dissipated by a straight apron or a **Morris and Johnson outlet** (preferred on larger structures — shorter apron).

Appurtenance	Role
Transverse sill	At the toe of the basin, induces the jump
Longitudinal sills	Straighten flow and protect side slopes
Chute blocks	At the basin entrance: stabilise the jump, raise effective depth, split flow into jets, create turbulence, lift jets off the apron, shorten the basin
Floor (friction, control) blocks	On the apron: dissipate energy, prevent scour, fix the jump position; two staggered rows, the second facing the gaps of the first
End sill	Low wall at the downstream end — vertical, stepped, sloped or dentated; rectangular, triangular or trapezoidal — reduces velocity and bed erosion
Baffle wall	Cross wall or vanes in a conduit that reduce eddies, even out velocity and dissipate energy
Training wall	Side walls that contain the flow with freeboard for spray and air entrainment
SAF basin	Designed at the Saint Anthony Falls Hydraulic Laboratory — chute blocks, floor blocks and end sill

Basin choice by Froude number

F_n	Behaviour	Basin
< 1.7	Conjugate depth $d_2 \approx 2d_1$, about 40% above critical depth; $V_2 \approx ½V_1$, about 30% below critical velocity	No special basin; channel length $\approx 4d_2$ beyond the depth-change point; energy devices usually unnecessary
1.7 – 2.5	Pre-jump stage, weak jump, no active turbulence	Baffles or sills usually unnecessary; basin long enough to contain the retarding flow
2.5 – 4.5	Transition stage, oscillating jump — the least effective dissipation; waves persist beyond the basin	Use the special basin of Fig. 16.13 (oscillations cannot be fully damped); adjust basin width to shift F_n
> 4.5	Fully developed jump	Blocks, baffles and sills shorten the basin and guard against sweep-out with low tailwater. Incoming velocity limited to 15–18 m/s (impact forces, cavitation); impact baffle blocks $0.8d_2$ from the start of the jump. Above 18 m/s omit impact baffles and rely on the jump — a longer basin; chute blocks and a dentated end sill shorten it

Trapezoidal basins may be cheaper with sloped side lining, but the jump is less complete and less stable than in rectangular basins — keep side walls vertical or nearly so.

Sources: ICAR eCourse 'Soil and Water Conservation Engineering' — Dr A. Mishra, reviewed by Dr B. C. Mal (IIT Kharagpur); AgriMoon e-book. US Bureau of Reclamation, Design of Small Dams and Hydraulic Design of Stilling Basins and Energy Dissipators (EM 25). Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. Flow with Froude number greater than 1 is

(a) tranquil (b) critical
(c) supercritical (d) laminar

Q2. Hydraulic depth is

(a) area/wetted perimeter (b) area/top width
(c) depth × width (d) velocity head

Q3. At critical depth in a rectangular channel, specific energy equals

(a) y (b) 1.5y
(c) 2y (d) 3y

Q4. Sequent depth ratio d_2/d_1 =

(a) $\sqrt{(1 + 8F^2)}$ (b) $½(\sqrt{(1 + 8F^2)} - 1)$
(c) $(1 + 8F^2)/2$ (d) 2F

Q5. An oscillating jump occurs for Froude numbers of

(a) 1–1.7 (b) 1.7–2.5
(c) 2.5–4.5 (d) 4.5–9

Q6. A steady, well-balanced jump occurs for F_n of

(a) 1.7–2.5 (b) 2.5–4.5
(c) 4.5–9 (d) > 9

Q7. Impact baffle blocks are placed from the start of the jump at about

(a) $0.2d_2$ (b) $0.8d_2$

(c) $2d_2$ (d) $4d_2$

Q8. Impact baffles are omitted when incoming velocity exceeds about

(a) 5 m/s (b) 10 m/s

(c) 18 m/s (d) 30 m/s

Answers

Q1 — (c) supercritical

Q2 — (b) area/top width

Q3 — (b) 1.5y

Q4 — (b) $\frac{1}{2}(\sqrt{(1 + 8F^2)} - 1)$

Q5 — (c) 2.5–4.5

Q6 — (c) 4.5–9

Q7 — (b) $0.8d_2$

Q8 — (c) 18 m/s